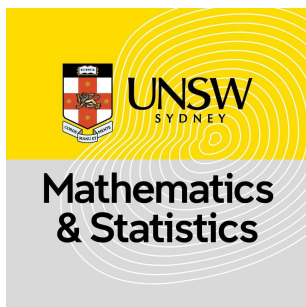




MATH3871/MATH5960 Bayesian Inference and Computation

Term 3, 2026

Tutorial Problems: Week 3

These exercises give further practice in computing posterior distributions, Jeffreys' priors, and conjugate analyses.

1. The quality of an electrical component is either excellent ($x = 1$), good ($x = 2$) or poor ($x = 3$). The probability of the various levels of quality depend on the factory of production ($\theta_1 = 0$: factory A; $\theta_1 = 1$: Factory B) and machine type ($\theta_2 = 0$: machine I; $\theta_2 = 1$: machine II; $\theta_2 = 3$: machine III). The conditional probabilities of $x = 3$ (likelihood) are in Table 1, and the joint prior distribution of (θ_1, θ_2) is in Table 2.

$\Pr(x = 3 \theta_1, \theta_2)$	$\theta_1 = 0$	$\theta_1 = 1$
$\theta_2 = 0$	0.2	0.3
$\theta_2 = 1$	0.4	0.1
$\theta_2 = 2$	0.5	0.2

Table 1: Conditional probabilities of getting a poor quality component ($x = 3$) given factory (θ_1) and machine type (θ_2).

$\Pr(\theta_1, \theta_2)$	$\theta_1 = 0$	$\theta_1 = 1$
$\theta_2 = 0$	0.1	0.2
$\theta_2 = 1$	0.2	0.3
$\theta_2 = 2$	0.1	0.1

Table 2: Prior probabilities of machine/factory combination for a randomly selected component.

- (a) Find the joint posterior distribution of $\theta_1, \theta_2|x = 3$, the posterior distribution over factory/machine combinations given that we have seen a low quality component, and each of its marginal distributions, $\theta_1|x = 3$ and $\theta_2|x = 3$.
 - (b) Has having observed $x = 3$ affected which factory/machine combination is the most likely to have produced the component?
2. Suppose x has the Pareto distribution $\text{Pareto}(a_0, b)$, where a_0 is known and fixed, but $b > 0$ is unknown, so that

$$f(x|b) = ba_0^b x^{-b-1} \quad \text{for } x > a_0.$$

Find Jeffreys' prior, and the corresponding posterior distribution for b when observing $x_1, \dots, x_n$.

3. Suppose that $x_1, \dots, x_n \sim N(\mu_0, \tau^{-1})$, where $\tau = 1/\sigma^2$ is the precision parameter, and where μ_0 is fixed. Suppose that we specify the prior $\tau \sim \text{Gamma}(\alpha, \beta)$.

- (a) Given that $\tau = 1/\sigma^2$, and that $\tau \sim \text{Gamma}(\alpha, \beta)$, show that σ^2 follows an Inverse Gamma distribution. The pdf of an $\text{InvGamma}(a, b)$ distribution is

$$f(x|a, b) = \frac{b^a}{\Gamma(a)} \left(\frac{1}{x}\right)^{a+1} \exp\left(-\frac{b}{x}\right) \quad x > 0, a > 0, b > 0.$$

- (b) Show that the $\text{Gamma}(\alpha, \beta)$ distribution is conjugate for the posterior of τ .

- (c) Show that the $\text{InvGamma}(a, b)$ distribution is conjugate for the posterior of σ^2 .

4. Let $x_1, \dots, x_n \in \mathbb{R}^d$ be n i.i.d. d -dimensional vectors. Suppose that we wish to model $x_i \sim N_d(\mu, \Sigma)$ for $i = 1, \dots, n$ where $\mu \in \mathbb{R}^d$ is an unknown mean vector, and Σ is a known positive definite covariance matrix.

- (a) Adopting the conjugate prior $\mu \sim N_d(\mu_0, \Sigma_0)$, where μ_0 and Σ_0 are known, show that the resulting posterior distribution for $\mu|x_1, \dots, x_n$ is $N_d(\hat{\mu}, \hat{\Sigma})$ where

$$\hat{\mu} = (\Sigma_0^{-1} + n\Sigma^{-1})^{-1}(\Sigma_0^{-1}\mu_0 + n\Sigma^{-1}\bar{x})$$

and

$$\hat{\Sigma} = (\Sigma_0^{-1} + n\Sigma^{-1})^{-1}.$$

- (b) Confirm the same conjugate prior analysis for the precision matrix. That is, suppose that $x_i \sim N_d(\mu, \Sigma)$ with known μ where $\Lambda = \Sigma^{-1}$. Let $\Lambda \sim W_d(\Lambda; v, A)$ follow a Wishart prior distribution, where the density function of a $W_d(\Lambda; v, A)$ random variable is:

$$f(\Lambda; v, A) = \frac{|A|^v}{\Gamma_d(v)} |\Lambda|^{v-(d+1)/2} \exp[-\text{tr}(A\Lambda)],$$

where $v > (d-1)/2$ is the degree of freedom, and $A \in \mathcal{R}^{d \times d}$ is a positive semi-definite matrix. Show that the posterior $\pi(\Lambda|x)$ is also a Wishart distribution with modified parameters.